

QUANTUM NUMBER

01 UNCERTAINTY PRINCIPLE

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \cdot m \Delta v \geq \frac{h}{4\pi}$$

Q. According to Heisenberg's uncertainty principle, $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$ which of the following is correct ?

a) If $\Delta x = 0$ then $\Delta p = \infty$
 b) If $\Delta v = 0$ then $\Delta p = 0$
 c) If $\Delta p = 0$ then $\Delta x = \infty$
 d) All are correct

Q. Find uncertainty in velocity, if uncertainty position is equal to uncertainty in momentum.

a) $\frac{h}{2\sqrt{\pi m}}$ b) $\frac{1}{2m} \sqrt{\frac{h}{\pi}}$ c) $\frac{1}{m} \sqrt{\frac{h}{\pi}}$ d) $\frac{1}{2} \sqrt{\frac{h}{m\pi}}$

Q. The uncertainty involved in the measurement of velocity within a distance of $0.1 \text{ \AA}$ is:

a) $5.79 \times 10^6 \text{ m/s}$ b) $5.79 \times 10^7 \text{ m/s}$
 c) $5.79 \times 10^8 \text{ m/s}$ d) $5.79 \times 10^9 \text{ m/s}$

Angular momentum

in n^{th} orbital
 $= mvr = n\hbar$

Orbital angular momentum

$$= \sqrt{l(l+1)} \hbar$$

Spin angular momentum

$$= \sqrt{s(s+1)} \hbar$$

PRINCIPLE QUANTUM NUMBER

In n^{th} Shell ,
 Number of subshells = n
 Number of orbitals = n^2
 Max. number of electrons = $2n^2$

Q. Find angular momentum of
 (i) 2s orbital (ii) 3d orbital
 (iii) 4p orbital (iv) e^- in 4th orbit

1 It describes shell or orbit
 $n = 1, 2, 3, 4, \dots$
 K, L, M, N, $\dots$

2 It describes size & energy of shell.
 $r \propto n^2$ $E \propto \frac{1}{n^2}$

3 It defines the angular momentum
 $mvr = \frac{nh}{2\pi}$

Q. Find maximum no. of e^- having
 (i) $n=4, s=-\frac{1}{2}$ (ii) $n=3, l=1, m=0$
 (iii) $n=2, l=0$ (iv) $n=3, l=1$

AZIMUTHAL QUANTUM NUMBER

1 It describes subshell value from 0 to $n-1$
 $l=0 \rightarrow s$ $l=2 \rightarrow d$
 $l=1 \rightarrow p$ $l=3 \rightarrow f$

2 Orbital angular momentum
 $= \sqrt{l(l+1)} \hbar, \hbar = \frac{h}{2\pi}$

3 Maximum no. of orbital in a subshell = $2l+1$
 Maximum no. of electrons in a subshell = $4l+2$

If $l=2$
 1) Orbital = d
 2) No. of orbitals = $2(2+1)=5$
 ($d_{xy}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{z^2}$)
 3) Total $e^-s = 2(2l+1) = 10 e^-s$
 4) Orbital angular momentum =
 $= \sqrt{2(2+1)} \hbar = \sqrt{6} \hbar$

Q. Which of the following set of quantum numbers is correct?

n	l	m	s
1) 4	0	0	$+\frac{1}{2}$
2) 5	2	3	$-\frac{1}{2}$
3) 2	-1	0	$+\frac{1}{2}$
4) 6	3	0	$-\frac{1}{2}$

MAGNETIC QUANTUM NUMBER
 Value of $m = -l \leq m \leq l$
 Total values of $m = 2l+1$

$n=4$
 $l=0$ $m=0$
 $l=1$ $m=-1, 0, +1$
 $l=2$ $m=-2, -1, 0, +1, +2$
 $l=3$ $m=-3, -2, -1, 0, +1, +2, +3$

SPIN QUANTUM NUMBER

SPIN
 — CLOCKWISE ($+\frac{1}{2}$)
 — ANTICLOCKWISE ($-\frac{1}{2}$)

STRUCTURE OF ATOM

ENERGY OF ORBITALS

1) Mono electronic species
 Energy defined upon n

1s < 2s = 2p < 3s = 3p = 3d

2) Multi electronic species
 3s < 3p < 4s < 3d

(n+l) rule

$\rightarrow$ As $(n+l) \uparrow, E \uparrow$

$\rightarrow$ If $(n+l)$ is same, then $n \uparrow, E \uparrow$

Orbital	2s	3d
(n+l) value	$n=2$ $l=0$ $n+l=2$	$n=3$ $l=2$ $n+l=5$

SHAPE OF ORBITALS

- 1) s orbital - Spherical shape
- 2) p orbital - dumb bell shape
- 3) d orbital - double dumb bell shape

NODES

$\Psi \rightarrow e^-$ wave function

$\Psi^2 \rightarrow$ probability of finding the electrons

* Node $\rightarrow$ Probability of finding the electron is zero.

* Node plane $\rightarrow$ Plane; where $\Psi^2 = 0$

* Radial nodes $\rightarrow n-l-1$

* Angular nodes = l

* Total nodes = $n-1$

FILLING OF ATOMIC ORBITAL

n=1
 n=2
 n=3
 n=4
 n=5
 n=6
 n=7
 n=8

Aufbau principle

Electrons are filled in the increasing order of energy
 $1s < 2s < 2p < 3s < 3p < 4s < 3d \dots$

Pauli's exclusion principle

No two electrons can have same four quantum numbers
 $1s^3$ - against Pauli's exclusion principle

Hund's rule

Pairing is only takes place after each orbital is singly occupied.
 $\uparrow \downarrow \uparrow \uparrow$ - Against Hund's rule



PHYSICS WALLAH